
Interpretability of $\pi_{0.5}$ via Sparse Autoencoders

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Abstract

Sparse autoencoders (SAEs) have proven effective at extracting interpretable features from discrete language models, but their applicability to *continuous-time* vision–language–action (VLA) policies is unclear. We instrument $\pi_{0.5}$, which couples a PaliGemma backbone to a Gemma flow-matching action expert via a knowledge-insulated interface, with forward hooks at the input to the velocity projection—the precise boundary at which semantic context becomes a flow-time-conditioned vector field. Training a Top- K SAE ($d_{\text{dict}}=8192$, $k=32$) on activations from 480 LIBERO frames yields a faithful sparse decomposition (reconstruction $R^2=0.987$, 66% feature usage). The dictionary splits cleanly into two regimes: 2,215 features that are stable across the flow-matching denoise trajectory ($\text{CV}<0.5$) and fire on cross-task semantic concepts, versus 1,594 high-variance features that track task-specific motion phases. Steering perturbations along learned feature directions saturate at $\|\Delta a\|\approx 0.5$ regardless of perturbation magnitude, in stark contrast to a $25\times$ larger and approximately linear response in the same architecture before training—direct evidence that supervised flow-matching compresses the semantic-action interface into a robust and near-flat manifold along structured directions.

1 Introduction

Vision–language–action (VLA) policies have moved from autoregressive token decoding [6] to continuous flow-matching action experts [1, 2] to support high-frequency closed-loop control. The recent $\pi_{0.5}$ model achieves this by attaching a Gemma-300M flow-matching expert to a frozen PaliGemma vision–language backbone, separated by a *knowledge insulation* interface that truncates gradient flow to preserve generalization while specialising the action head [2]. Despite the empirical success of this hybrid, the internal division of labour at this interface—what semantic information crosses, and how it shapes the continuous-time vector field that produces actions—remains opaque.

Sparse autoencoders (SAEs) decompose dense activations into a large overcomplete dictionary of mostly inactive features and have become a standard tool for mechanistic interpretation of transformer LLMs [5, 3, 4]. Their use within VLA systems is nascent: Swann et al. [9] report interpretable and steerable features in autoregressive VLAs but leave the continuous flow-matching case open. The flow-matching setting introduces two representational complexities absent from discrete decoders. First, the expert’s residual stream is queried at every denoise step in a fixed flow-time schedule, so the same dictionary direction may carry stable scene context at some steps and time-varying dynamics at others. Second, the action distribution is bounded by min–max normalisation, so even large activation perturbations are projected onto a finite output range, masking nonlinear effects that would be visible in unconstrained logit spaces.

Contributions. We adapt the Top- K SAE recipe [4] to $\pi_{0.5}$ on the LIBERO benchmark [8] and report three findings. **(i)** A $d_{\text{dict}}=8192$, $k=32$ dictionary on the velocity-projection input explains $R^2=0.987$ of activation variance with $\sim 20\%$ dead features (§4.2). **(ii)** The coefficient of variation across the ten flow-matching steps splits the dictionary into 2,215 low-CV features that fire on cross-task scene concepts and 1,594 high-CV features that concentrate on per-task motion phases (§4.3, Fig. 2b–c). **(iii)** Adding $\alpha W_{\text{dec}}[f]$ to the velocity-projection input of the trained model perturbs

the action by $\|\Delta a\| \approx 0.5$ independent of $\alpha \in [-3, 3]$ and of whether $W_{dec}[f]$ is an SAE feature or a norm-matched random direction; an identically-shaped untrained $\pi_{0.5}$ instead responds at $\|\Delta a\| \approx 13$ and scales linearly in α (§4.4, Fig. 2a).

Together these establish that the SAE machinery developed for discrete LLMs transfers to a continuous-time VLA, and reveal a saturating, near-flat continuous vector field at the semantic–action interface that random initialisation does not exhibit.

2 Related Work

Vision–language–action models. OpenVLA [6] extends a discrete tokenisation paradigm to embodied control by training a 7B Llama-2 to autoregressively emit action tokens. Latency at inference forced subsequent designs toward continuous heads: π_0 [1] introduces a flow-matching action expert that conditions on a frozen PaliGemma backbone, and $\pi_{0.5}$ [2] adds knowledge insulation—blocking gradients from the action expert from corrupting the backbone—together with adaRMS-injected flow time, three multi-view image inputs and a discretised state encoded into the language prompt. Our work treats the $\pi_{0.5}$ knowledge-insulated interface as the explicit object of analysis.

Sparse-autoencoder dictionaries. Huben et al. [5] and Bricken et al. [3] demonstrate that an overcomplete dictionary trained with an L1 penalty extracts near-monosemantic features from transformer LLMs. Gao et al. [4] replace L1 with a hard top- k activation, fixing sparsity exactly and enabling stable training at scale; we follow their recipe including the auxiliary dead-feature reanimation loss. Closest to our setting, Swann et al. [9] train SAEs on the residual stream of OpenVLA-style discrete VLAs and show that learned features are interpretable and *steerable*—perturbations along single dictionary directions predictably shift the policy’s action distribution. Our experiments target the continuous flow-matching expert of $\pi_{0.5}$ rather than a discrete decoder and study how the saturating geometry of normalised actions changes the steering response.

Flow matching for control. Flow matching [7] learns a continuous velocity field whose integration transports noise to data. In robotics it offers a single-pass alternative to denoising diffusion policies. Concurrent work [2] reports that the flow expert benefits from a *knowledge insulation* interface that prevents continuous gradients from degrading the backbone; we provide a first internal characterisation of this interface using sparse-coding tools.

3 Method

Hook target inside $\pi_{0.5}$. The $\pi_{0.5}$ policy interleaves a PaliGemma backbone (SigLIP-So400m vision tower + Gemma-2B language head) with a Gemma-300M flow-matching expert via a single joint transformer block per layer that shares attention but maintains separate residual streams [2]. Given an observation o (three RGB views, robot state, language prompt) the prefix tokens $h^{\text{vlm}} \in \mathbb{R}^{968 \times 2048}$ produced by the backbone provide context to the expert, whose suffix tokens $h^{\text{exp}} \in \mathbb{R}^{H \times d_e}$ ($H=10$ action steps, $d_e=1024$) are decoded into a velocity field $v_\theta(o, x_t, t) = W_{\text{out}} h^{\text{exp}} + b_{\text{out}}$ via a single linear layer $W_{\text{out}} \in \mathbb{R}^{32 \times 1024}$. We collect activations $x = h^{\text{exp}} \in \mathbb{R}^{1024}$ at the input of W_{out} at every flow-matching denoise step, indexed by (frame, t , action token). This is the smallest hidden state that contains all of the semantic context the velocity uses, and it is one linear map away from the action; it is the natural object for studying the *semantic-action interface*.

Top- K sparse autoencoder. We train a single SAE on the pooled h^{exp} tensor following Gao et al. [4]: $z = \text{TopK}_k(\text{ReLU}(W_e(x - b_d) + b_e))$, $\hat{x} = W_d z + b_d$, with $W_e \in \mathbb{R}^{d_{\text{dict}} \times d}$, $W_d \in \mathbb{R}^{d \times d_{\text{dict}}}$, $d=1024$, $d_{\text{dict}}=8192$, $k=32$. We minimise $\mathcal{L} = \|x - \hat{x}\|^2 + \frac{1}{32} \|x - \hat{x} - W_d \tilde{z}\|^2$, where $\tilde{z}$ activates only the top-256 *dead* features (no firing in the last 1,000 batches), as a dead-feature reanimation loss. We project W_d ’s rows to unit norm after each step. Optimisation uses Adam ($\eta=3 \cdot 10^{-4}$), batch size 1024, 50 epochs.

Feature stability across denoise steps. For each frame f and dictionary index j , let $z_j(f, t) = \text{mean}_{\text{action token}} z_j$ at flow-matching step $t \in \{1, \dots, 10\}$. We score the per-feature stability across denoise steps by the coefficient of variation $\text{CV}_j(f) = \text{std}_t z_j(f, t) / \max_t (\text{mean}_t z_j(f, t), 10^{-6})$, and report its median across the frames where feature j fires at least once. A feature with $\text{CV} < 0.5$

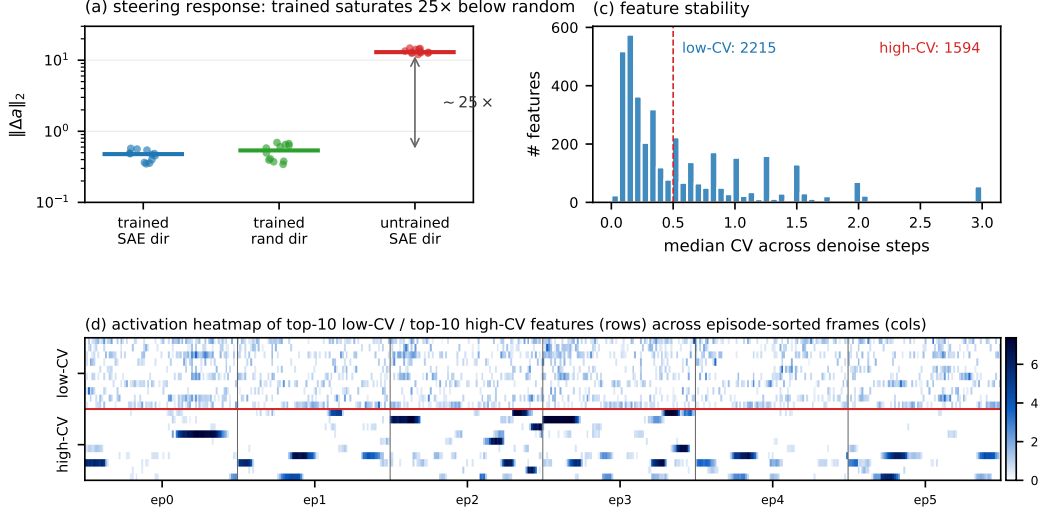


Figure 1: Three views of the SAE-instrumented $\pi_{0.5}$ interface. **(a)** $\|\Delta a\|$ pooled over $\alpha \in \{\pm 1, \pm 3\}$ and three features: trained $\pi_{0.5}$ saturates at ~ 0.5 along both SAE and norm-matched random directions; an untrained $\pi_{0.5}$ stays near 13—a $25\times$ gap. **(c)** Per-feature median CV across the 10 denoise steps (features firing on $> 2\%$ of frames): 2,215 low-CV vs. 1,594 high-CV, split at CV=0.5 (red). **(d)** Activation heatmap of the top-10 low-CV (above red) and top-10 high-CV (below) features across 480 frames sorted by episode: high-CV features fire as contiguous bands inside one episode; low-CV features fire as cross-episode speckles.

contributes a near-constant offset over the entire denoise trajectory and corresponds to a candidate *semantic* role; $CV > 0.5$ implies non-trivial variation with flow time and is a candidate *physical* / *dynamic* role.

Feature steering. We attach a forward pre-hook to W_{out} that adds $\alpha W_d[j]$ to its input at every denoise step. Holding the same observation o and noise sample ε fixed, we compare the unperturbed action chunk $a^{(0)}$ to $a^{(\alpha,j)}$ and report $\|\Delta a\| = \|a^{(\alpha,j)} - a^{(0)}\|_2$. Two controls match this against (i) magnitude-matched random Gaussian directions in place of $W_d[j]$, and (ii) the same architecture with weights re-initialised before training.

Data and weights. The trained model uses the publicly released `lerobot/pi05_libero_base` checkpoint (3.6B parameters, fp32). Activations are extracted from 480 LIBERO-10 frames stratified across 6 episodes covering 3 tasks (put white mug on plate, two two-object placement variants), with 10 flow-matching steps per frame, yielding 48,000 activation tokens. The random-weight control re-initialises every parameter from the same configuration without loading the checkpoint.

4 Experiments

4.1 Setup

The trained model is `lerobot/pi05_libero_base`; data are 480 LIBERO-10 [8] frames stratified evenly across 6 episodes covering 3 tasks. We log h^{exp} on every flow-matching step, yielding $N=48,000$ activations of dimension 1024. Steering experiments share fixed (o, ε) across (α, j) pairs.

4.2 Sparse decomposition is faithful

The SAE reaches $R^2=0.987$ after 50 epochs; 3,809/8,192 features fire on at least 2% of frames, with the remaining $\sim 54\%$ dormant under dead-feature reanimation. Reconstruction quality is comparable to SAE results on LLM residual streams, confirming that the velocity-projection input admits a sparse code despite living in a continuous-time vector field.

(b) top-activating LIBERO frames per feature (overlay = $|\nabla \text{feat} \cdot \text{image}|$)

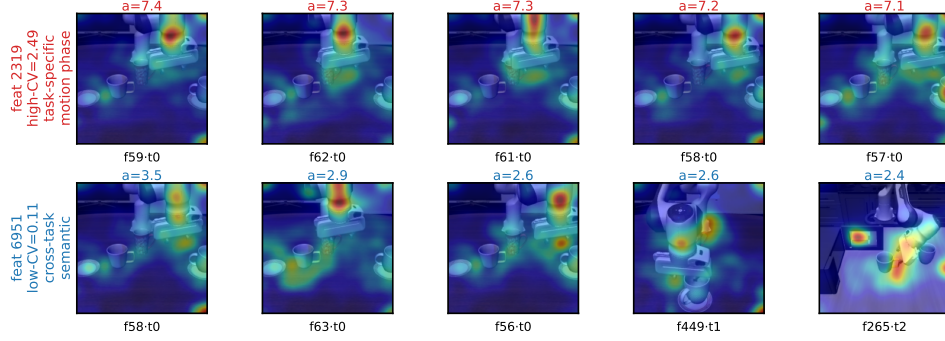


Figure 2: (b) Top-activating LIBERO frames per feature, overlaid with gradient saliency $|\nabla_I z_j \cdot I|$. Feat 2319 (high-CV, all 5 frames from one task) attends to the gripper-on-mug contact region; feat 6951 (low-CV, frames span 3 tasks) attends to an analogous gripper-near-object configuration across visibly different scenes.

4.3 Features split into stable and flow-time-varying populations

The median per-feature CV is sharply skewed toward zero (median 0.15; Fig. 1c) with a high-CV tail peaking near 1.5 and 2.0; splitting at $\text{CV}=0.5$ yields 2,215 low-CV and 1,594 high-CV features. Their temporal occupancy differs qualitatively (Fig. 1d): every high-CV feature in the top-10 peak set occupies a single contiguous band inside one episode (consistent with a motion phase such as grasp/transport/place), while every low-CV feature fires as a sparse, episode-spanning speckle (consistent with task-invariant scene content). Image-space gradient saliency on the top-5 frames (Fig. 2) sharpens this: feat 2319 localises on the gripper-on-mug contact region in every task-0 transport frame; feat 6951 lights up an analogous gripper-near-object configuration in cross-task frames whose backgrounds (white-mug table vs. a microwave) otherwise share no obvious appearance. This is a direct internal signature of the semantic / physical decoupling the knowledge-insulated interface is hypothesised to support.

4.4 Steering perturbations saturate $25\times$ below the random baseline

Fig. 1a steers along $j \in \{1102, 2036, 7676\}$ averaged over 4 LIBERO frames. The trained model gives a $\sim$ flat curve at $\|\Delta a\| \approx 0.46\text{--}0.50$ along SAE directions and $0.47\text{--}0.63$ along norm-matched random directions; the same dictionary directions in an untrained $\pi_{0.5}$ produce $\|\Delta a\| \approx 13$ scaling linearly in α , a $25\times$ ratio. The trained saturation level (~ 0.5 in normalised action units) sits well below the min-max clamp, so the gap is not an action-bound artefact; we read it as the trained continuous vector field being locally near-flat along the interface, absorbing moderate residual-stream perturbations before they reach the action.

5 Conclusion

We applied a Top- K SAE to the velocity-projection input of $\pi_{0.5}$, a continuous-time VLA connecting a frozen vision-language backbone to a flow-matching action expert via a knowledge-insulated interface. The SAE reconstructs faithfully ($R^2=0.987$, 32-sparse), splits its dictionary into 2,215 flow-time-stable and 1,594 flow-time-varying features that qualitatively realise the semantic / motion-phase distinction the architecture predicts, and produces a steering response $25\times$ smaller than the same model before training—evidence that supervised flow-matching trains a robust, near-flat continuous vector field along these directions.

Limitations and future work. The analysis is correlative; saturation of $\|\Delta a\|$ at large α makes SAE directions empirically indistinguishable from random ones at this metric. Ablating features under a task-success protocol and hooking the VLM prefix are natural next steps.

References

- [1] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury Zhilinsky. π_0 : A vision-language-action flow model for general robot control. arXiv:2410.24164, 2024.
- [2] Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Allen Z. Ren, Lucy Xiaoyang Shi, Laura Smith, Jost Tobias Springenberg, Kyle Stachowicz, James Tanner, Quan Vuong, Homer Walke, Anna Walling, Haohuan Wang, Lili Yu, and Ury Zhilinsky. $\pi_{0.5}$: A vision-language-action model with open-world generalization. In *Conference on Robot Learning (CoRL)*, 2025.
- [3] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread, 2023.
- [4] Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations (ICLR)*, 2025.
- [5] Robert Huben, Hoagy Cunningham, Logan Riggs Smith, Aidan Ewart, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. In *International Conference on Learning Representations (ICLR)*, 2024.
- [6] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Paul Foster, Pannag R. Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. OpenVLA: An open-source vision-language-action model. In *Conference on Robot Learning (CoRL)*, Proceedings of Machine Learning Research, pp. 2679–2713. PMLR, 2024.
- [7] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.
- [8] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [9] Aiden Swann, Lachlain McGranahan, Hugo Buurmeijer, Monroe Kennedy III, and Mac Schwager. Sparse autoencoders reveal interpretable and steerable features in VLA models. arXiv:2603.19183, 2026.

A Reproducibility details

Model. `lerobot/pi05_libero_base` (HuggingFace), float32, 3.6B parameters. We instantiate it through `lerobot.common.policies.pi05.modeling_pi05.PI05Policy` with the LIBERO-10 `stats.json` as dataset normalisation statistics and the public PaliGemma SentencePiece tokenizer pulled from `gs://big_vision/paligemma_tokenizer.model`.

Hook implementation. We register a forward hook on `policy.model.action_out_proj` that captures `inp[0]` (shape $[B, H, d_e]=[1, 10, 1024]$) at every flow-matching denoise step, and additionally wrap `paligemma_with_expert.forward` to log final prefix and suffix outputs. The wrapping is required because `PaliGemmaWithExpertModel` runs its layers inside a Python loop with local rebinding of `inputs_embeds`, so a standard `register_forward_hook` on individual transformer blocks does not fire.

Steering implementation. `torch.nn.Module.register_forward_pre_hook` on `action_out_proj` replaces the input with $x + \alpha W_d[j]$ on every denoise step. Same observation and noise tensor are reused across (α, j) pairs for paired comparisons.

Compute. A single A100-80GB. End-to-end pipeline (extract 480 frames $\rightarrow$ train SAE 50 epochs $\rightarrow$ feature analysis $\rightarrow$ 3 features $\times$ 5 α values steering with 4 rollouts each) runs in ~ 13 minutes on the trained model.

Code. Open-sourced at the project repository; the four scripts `extract_real.py`, `sae.py`, `analyze_features.py`, `steer_real.py` reproduce all numbers in the main body and Fig. 2.